USER MANUAL · BCA 8TH SEMESTER PROJECT

Loan Approval Prediction System

How to start the system, check whether a loan application would be approved, read your results, and manage banks, blogs and feedback as an administrator.

Model Decision Tree (scikit-learn) **Stack** Next.js · Express · Flask · MariaDB

Address `http://localhost:3000`

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1 · ABOUT

About the system

The system predicts whether a bank would approve a personal loan, based on the applicant's details. It was trained on a public loan dataset of 614 past applications. Users get an instant “Approved” or “Not Approved” answer, and every result is saved to their history. Administrators keep the bank rate list and blog up to date and read user feedback.

It runs as four parts that talk to each other:

`:3000`
Website
Next.js pages the user sees: forms, history, bank list, blog, admin panel.

`:5001`
Prediction service
Flask app that loads `model.pkl` and returns 1 (approve) or 0 (reject).

`:8080`
API
Express server for login, saving predictions, banks, blogs and feedback.

`:3306`
Database
MariaDB database `loanprediction` with users, predictions, banks, blogs, feedback.

When you submit the loan form, the website first asks the prediction service for an answer, then sends the answer to the API, which stores it in the database.

2 · SETUP

Starting the system

The database starts automatically with the computer. The other three parts each need their own Terminal window. Start them in this order, then open `http://localhost:3000`.

```
# Terminal 1: prediction service (port 5001)
cd ~/Desktop/eight_sem_project/Loan_Prediction
.venv-mac/bin/python app.py

# Terminal 2: API (port 8080)
cd ~/Desktop/eight_sem_project/loan_api
npm start

# Terminal 3: website (port 3000)
cd ~/Desktop/eight_sem_project/final_Project
npm run dev
```

KEEP ALL THREE RUNNING

If you close a Terminal window, that part stops. Without the prediction service the form gives no answer; without the API nobody can log in.

3 · APPLICANT

Using it as an applicant

1 Create an account

Click **Sign up** in the top menu. Enter your first name, last name, email and a password, type the password again to confirm, and submit.

2 Log in

Click **Login** and enter the same email and password. You return to the home page, now signed in.

3 Fill in the prediction form

Click **Predict** in the menu. If you aren't logged in, the site sends you to the login page first. Fill in every field. Section 4 explains each one.

4 Submit and read the answer

Click submit. A popup says **Loan Approved** or **Loan Not Approved**. The result is saved to your account automatically.

5 Look back at past results

Open the portal and choose **Prediction** in the sidebar (`/portal/predictresult`). Each row shows one application with its date and result. Click a row to see every value you entered.

6 Explore the rest of the site

- **Bank** compares personal loan interest rates, tenures and processing fees at Nepali banks such as NIC Asia, RBB and Nepal SBI.
- **Blog** has short articles about loans and borrowing.
- **Details** describes the system.
- **Feedback**: while logged in, send a message to the administrators from the home page.

4 · REFERENCE

Form fields

The model learned from applications recorded in a specific format, so enter numbers on the same scale as the examples.

FIELD	WHAT TO ENTER	EXAMPLE
Gender	Male or Female	Male
Married	Yes or No	Yes
Dependents	People who rely on your income, 0 to 4	0
Education	Graduated or Ungraduated	Graduated
Self Employed	Yes if you run your own business	No
Applicant Income	Your monthly income	5000
Coapplicant Income	Monthly income of a co-applicant, such as a spouse. Enter 0 if none.	2000
Loan Amount	Amount requested, **in thousands**	150 = 150,000
Loan Amount Term	Repayment period **in months**	360 = 30 years
Credit History	1 if you have repaid past loans and cards on time, 0 if not	1
Property Area	Where the property is: Urban, Semi Urban or Rural	Urban

5 · OUTPUT

Reading the result

EXAMPLE APPLICATION

Applicant Income	Coapplicant Income
5000	2000
Loan Amount	**Loan Amount Term**
150	360
Credit History	**Property Area**
1	Urban

`Credit History = 1`

Loan Approved

Congratulations! Your loan has been approved.

`Credit History = 0`

Loan Not Approved

Sorry, your loan has not been approved.

The result is a prediction, not a bank decision. It says whether applicants with similar details were approved in the training data. In your history it appears as **Yes** (approved) or **No** (not approved).

6 · ADMINISTRATOR

Using it as an admin

Log in on the normal **Login** page with an admin account. The site checks the account's role and opens the admin panel at `/admin/dashboard`.

EMAIL	PASSWORD
admin@gmail.com	Admin@123

The left sidebar has these sections:

MENU	WHAT YOU CAN DO
Dashboard	See a chart of how many applications were approved and how many were not.
Prediction	Browse every user's applications, open one to see all its values, or delete it.
Bank	Add a bank with its loan name, interest rate, tenure and processing fee, or edit and delete one. Changes appear on the public Bank page.
Blogs	Write a post with a title, text and image, or edit and delete one. Posts appear on the public Blog page.
Feedback	Read messages that users sent, with the sender and date.
Logout	Sign out and return to the login page.

SWITCHING ACCOUNTS

The browser remembers the last login. Log out before signing in as a different user, or use a private window for the second account.

7 · HELP

Troubleshooting

The form submits but no popup appears

The prediction service isn't running. Start Terminal 1 from section 2 and try again.

Login says the email or password is wrong

Check the spelling, or sign up again. If every login fails, the API (Terminal 2) or the database has stopped. Restart the database with `brew services start mariadb`.

The admin panel sends me back to the login page

The admin pages need a logged-in account. Log in on `/login` with the admin email and password first.

The prediction service says port 5000 is in use

macOS AirPlay Receiver uses port 5000. This project runs the service on 5001 instead, so make sure `app.py` starts with `port=5001`.

The API stops with a bcrypt error

Run `npm install bcrypt@6` inside `loan_api`, then start it again.

8 · LIMITATIONS

Known limitations

CREDIT HISTORY DECIDES THE RESULT

The trained decision tree has a single split on Credit History. Applicants with a value of 1 are approved and those with 0 are not, whatever their income or loan amount. A deeper tree or a random forest would let the other fields count.

Other limitations of the current version:

AREA	LIMITATION
Dependents	Collected and saved, but not sent to the model.
Encodings	The form's codes for Education and Property Area are in a different order from the ones used in training.
Access control	Adding, editing and deleting banks, blogs and predictions doesn't check for an admin login on the server.
Data	Training data comes from a public dataset, not from Nepali banks, so results shouldn't be used for real lending decisions.